b. Intermittent system

In this system, treated water is supplied to the consumers during certain fixed hours of the day. This system is used when the quantity of water is limited and the pressure in the system is low.

Merits

i. Minimum wastage due to vigilant use of consumers.
ii. Suitable if the source has only limited supply.
iii. Repairs can be easily attended during non-supply hours.
iv. Own storage of the consumers will relieve the situation during emergencies or non-supply.

Demerits

i. The consumers have to store water for use during non-supply hours.
ii. If emergencies such as fire occur during non-supply period, it causes great loss and hardship.
iii. More wastage due to open tap, etc.

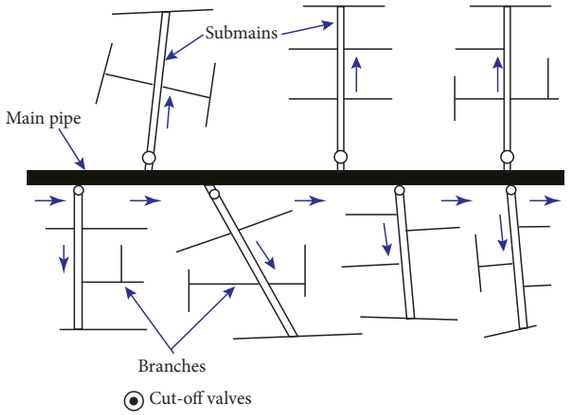
4.7.3 Methods of Layout of Distribution pipes

There are four systems of layout of pipes

1. Dead end or tree system.
2. Grid-iron system or interlaced system or reticulation system.
3. Circular or ring system.
4. Radial system.

1. Dead end or tree system

This system shown in fig, is more suitable to irregularly developed areas without any properly planned roads. This system consists of one supply main and from which many submains are taken. The submains again branch off into several branch lines. The branch lines divide into a number of service pipes.



Merits

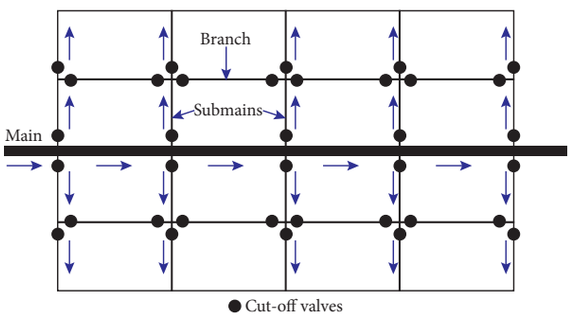
i. Cheap and simple and can be expanded, extended easily.
ii. Discharge and pressure calculations are easy, simple and accurate.
iii. Requires less number of cut off valves.
iv. Pipe lines can be laid in any pattern easily.

Demerits

i. During repairs, large portion of distribution area is affected.
ii. Water available for fire fighting will be limited.
iii. The numerous dead ends of the system prevent the free circulation of water.
iv. The stagnated water at dead ends may pollute.

2. Grid -Iron system

In this system shown in fig., the mains, submains and branches are all interconnected with each other. The dead ends are completely eliminated. This system is most suitable for well planned towns.



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